

Ramanathan Murugappan

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Gen AI Architect with 6+ years of experience building robust, end-to-end, enterprise-grade cross-platform AI products with high proficiency in ML Modeling and agentic frameworks.

PROFESSIONAL EXPERIENCE

AI/ML Lead Research Engineer (R&D) – ITC Infotech 	Mar'25 – Present
• HR RAG App: * Architected a high-performance RAG application for the HR knowledge base with 700+ documents using Docling for document parsing. * Engineered a multi-vector RAG pipeline with hybrid search (OpenSearch). * Integrated Agentic RAG capabilities via the Open WebUI frontend. * Built an eval and observability stack with DeepEval, LangSmith, and Langfuse. • ServiceNow Multi-Agent System: * Led development of a multi-agent architecture for end-to-end ServiceNow automation. * Implemented a Master Orchestrator Agent that leverages MCP for dynamic agent routing and execution.	
Data Science Analyst (Data & AI) – Accenture 	Aug'21 – Mar'25
• Retail Lens: * Engineered a visual search tool using SAM for background removal and Clip-ViT-B for image embeddings. * Integrated a Qdrant vector database, significantly improving user experience in visual product searches. • GENAI Asthma Prediction Tool: * Developed a GENAI asthma prediction tool with RAG and LLM chat functionality integrated with Excel/CSV. * Led front-end (Streamlit) and back-end development, contributing to two project phases and client demos. • Fee-Optimizing Model: * Engineered a fee-optimizing model for plasma donations using segmentation and profiling of customer data. * Utilized automated web scraping for data collection, enhancing the effectiveness of the pricing strategy.	
Data Science Analyst – Kaleidofin 	Dec'19 – Aug'21
• Credit Risk Model: * Developed a credit risk model using Bagging and Boosting to score new-to-credit and MFI customers. * Conducted monthly risk analyses for continuous improvements in code base performance. • Payment Prediction Model: * Built a payment prediction model using RandomForest, LightGBM, and GridSearchCV for hyperparameter tuning. * Improved efficiency and optimization of in-house call center operations by accurately predicting customer payments. • Dashboards & Database Automation: * Designed and deployed customized dashboards for partners and teams, automating workflows with Apache Airflow. * Built data pipelines to integrate various data sources, enriching dashboards with granular data.	

SKILLS

Gen AI & LLM	LangChain, LangGraph, LlamaIndex, LiteLLM, PydanticAI, Hugging Face, Ollama, OpenAI, Anthropic, Google Gemini, Groq, Mistral, Docling
Agentic AI	MCP, A2A, Tool Calling, ReAct Agents (Create agent), RAG, Multi-Agent, Agent Middleware (Summarization, ContextEditing, HumanInTheLoop, ModelFallback), Structured Output, Computer Use Agents
Fine-Tuning & Alignment	LoRA, QLoRA, RLHF, Instruction Tuning, DPO, PEFT, Unsloth, Axolotl, TRL
Model Modalities	Multimodal, Vision, Voice & Audio (ElevenLabs, Whisper, XTTS-v2), STT/TTS (Deepgram, AssemblyAI)
AI Dev Tools	Cursor, Windsurf, Claude Code, GitHub Copilot, Cline, Replit, Lovable, Bolt.new
Evaluation & Observability	LangSmith, LangFuse, Arize Phoenix, DeepEval, RAGAs, Weights & Biases, Guardrails AI
Vector DB	Qdrant, Pinecone, Weaviate, ChromaDB, LanceDB, FAISS, OpenSearch, Azure AI Search
Databases & ETL	PostgreSQL, MongoDB, DynamoDB, BigQuery, Apache Airflow, Apache Kafka, AWS RDS
Languages	Python, SQL, JavaScript
Frameworks	FastAPI, React.js, Streamlit, Gradio, Pandas, Apache Spark
MLOps	Docker, MLflow, Git, OpenShift, Podman
Cloud & Infra	AWS (Lambda, EC2, S3, Bedrock, VPC), GCP, Azure, Modal, Together AI, Bash Scripting

ACHIEVEMENTS & OTHER PROJECTS

Personal Website – ram96.com

- Designed and built a personal portfolio website showcasing projects, research, and professional experience.
- Serves as a central hub for my **digital presence**, integrating the Resume Chatbot (Digital Clone) and project highlights.

GrowthX – LinkedIn Post

- Developed a growth strategy to scale Blue Tokai's revenue from **₹250 crore to ₹500 crore within 12 months**.
- Won the Capstone by presenting to **1,000+ industry professionals**, with founders expressing interest in **80% of recommendations**.

Resume Chatbot

- Built a web-based chatbot to simulate resume-related interview questions using **Gemini (LLM)**.
- Deployed with **Flask** and **React.js**, containerized with **Docker**, and hosted on **Render**.

Flight Delay Prediction

- Built a two-stage ML engine to forecast on-time performance of US flights using weather data.
- First stage performs binary **classification** for delay prediction, second stage uses **regression** to predict delay duration in minutes.

PUBLICATIONS

SOLARILLION FOUNDATION (SF)

Aug'18 – May'20

Research Assistant + Teaching Assistant

A Two-Stage Machine Learning Approach to Forecast the Lifetime of Movies in a Multiplex  at Future of Information and Communication Conference (FICC) 2020, San Francisco, USA.

User-Independent Human Stress Detection  at IEEE Intelligent Systems IS'20 - Varna, Bulgaria, August 2020.

EDUCATION

M.E. Mechatronics

2018-2020

Anna University (M.I.T Campus)

Chennai

CERTIFICATIONS

Red Hat Certified Specialist in OpenShift Administration 

Advanced Analytics for Data Scientists by Workera 

Generative AI for Developers by Google 

Responsible AI by Workera 